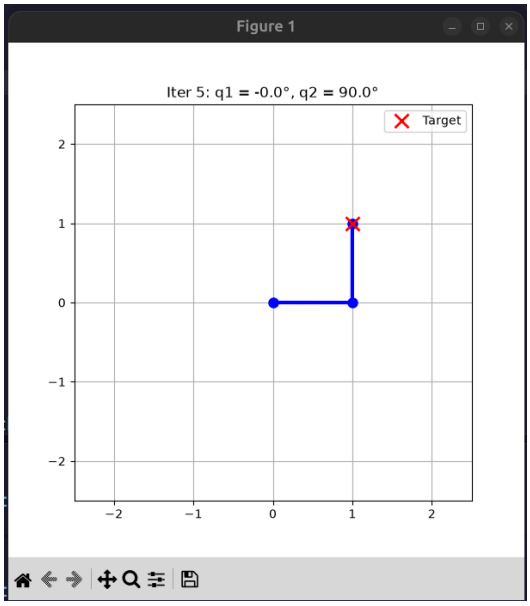
EE430 Laboratory Report 1: Jacobian and Velocity Kinematics

Part 1: Numerical Inverse Kinematics using RR Robot

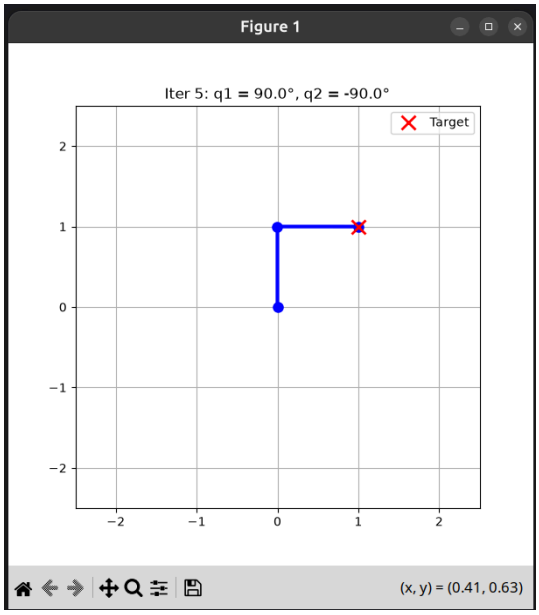
Output:-



Initial Guess of 2, 2 radians

```
1 to choose target coordinate, 2 to choose target joint angles
1
Input target co-ordinate in x,y format or target angle in q1,q2 format (in degrees)
1,1

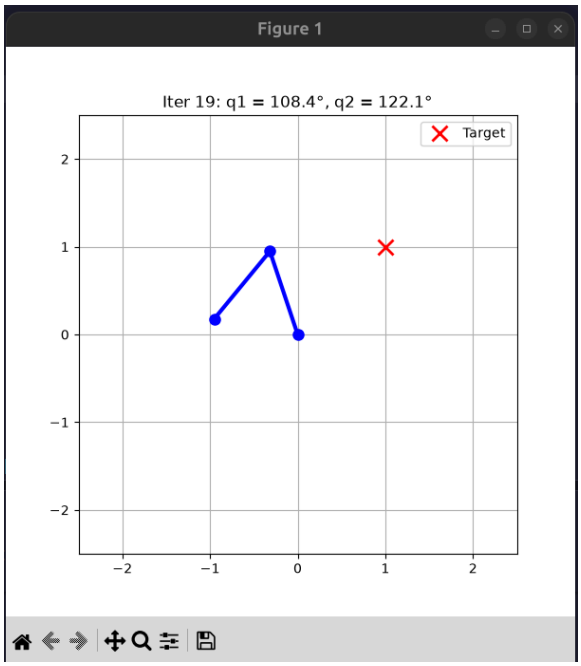
Converged in 5 iterations!
Joint Angles: q1 = -0.04°, q2 = 90.04°
```



Initial Guess of 10, 10 radians

```
1 to choose target coordinate, 2 to choose target joint angles
1,1
Invalid entry. Try again.
1 to choose target coordinate, 2 to choose target joint angles
1
Input target co-ordinate in x,y format or target angle in q1,q2 format (in degrees)
1,1

Converged in 5 iterations!
Joint Angles: q1 = 90.01°, q2 = -90.02°
```



Initial guess of 0,1.57 radians. Singularity.

Part 2: Derivation of a Jacobian

Output:-

```
[-3.45625683e-01 -4.89897949e-01  5.00000000e-01 -1.56066017e-01]
 [ 7.52231083e-01 -2.82842712e-01 -8.66025404e-01  2.53652968e-02]
 [ 0.00000000e+00  4.24264069e-01  6.12323400e-17 -1.22474487e-01]
 [ 0.00000000e+00  5.00000000e-01  0.00000000e+00 -6.12372436e-01]
 [ 0.00000000e+00 -8.66025404e-01  0.00000000e+00 -3.53553391e-01]
 [ 1.00000000e+00  6.12323400e-17  0.00000000e+00  7.07106781e-01]]
```

Analytical verification:

The robot has 4 degrees of freedom configured in an RRPR architecture (Revolute, Revolute, Prismatic, Revolute). **Joint Variables:** $q_1 = \theta_1$, $q_2 = \theta_2$, $q_3 = d_3$, $q_4 = \theta_4$

Denavit-Hartenberg Parameters

Joint (i)	Type	a_i (m)	α_i (deg)	d_i (m)	Given State
1	Revolute	0.40	90°	0.30	30°
2	Revolute	0.60	0°	0.00	45°
3	Prismatic	0.00	-90°	0.25 m	0°
4	Revolute	0.20	90°	0.10	60°

The standard Denavit-Hartenberg transformation matrix T_{i-1}^{i} is defined as:

```
[ cos(θi)   -sin(θi)cos(αi)    sin(θi)sin(αi)   ai*cos(θi) ]
[ sin(θi)    cos(θi)cos(αi)   -cos(θi)sin(αi)   ai*sin(θi) ]
[ 0          sin(αi)          cos(αi)          di         ]
[ 0          0                0                1          ]
```

Substituting the parameters for each joint:

$T_0^1(q1)$ =

```
[ cos(q1)   0    sin(q1)   0.4*cos(q1) ]
[ sin(q1)   0   -cos(q1)   0.4*sin(q1) ]
[ 0         1    0         0.3         ]
[ 0         0    0         1           ]
```

$T_1^2(q2)$ =

```
[ cos(q2) -sin(q2)   0   0.6*cos(q2) ]
[ sin(q2)  cos(q2)   0   0.6*sin(q2) ]
[ 0        0         1   0           ]
[ 0        0         0   1           ]
```

$T_2^3(q3)$ =

```
[ 1   0   0   0   ]
[ 0   0   1   0   ]
[ 0  -1   0   q3  ]
[ 0   0   0   1   ]
```

$T_3^4(q4)$ =

```
[ cos(q4)   0    sin(q4)   0.2*cos(q4) ]
[ sin(q4)   0   -cos(q4)   0.2*sin(q4) ]
[ 0         1    0         0.1         ]
[ 0         0    0         1           ]
```

For Revolute Joints (i = 1, 2, 4):
Linear velocity: $J_{vi} = z_{i-1} \times (o_4 - o_{i-1})$

Angular velocity: $J_{wi} = z_{i-1}$
For Prismatic Joint (i = 3):
Linear velocity: $J_{v3} = z_2$
Angular velocity: $J_{w3} = [0, 0, 0]^T$

q1 = 30°, q2 = 45°, q3 = 0.25 m, q4 = 60°

o_0 = [0.0000, 0.0000, 0.0000]T
z_0 = [0.0000, 0.0000, 1.0000]T

o_1 = [0.3464, 0.2000, 0.3000]T
z_1 = [0.5000, -0.8660, 0.0000]T

o_2 = [0.7138, 0.4121, 0.7243]T
z_2 = [0.5000, -0.8660, 0.0000]T

o_3 = [0.8388, 0.1956, 0.7243]T
z_3 = [-0.6124, -0.3536, 0.7071]T

o_4 = [0.7522, 0.3456, 0.8657]T (End-effector position)
J_v1 = z_0 × (o_4 - o_0) = [-0.3456, 0.7522, 0.0000]T
J_w1 = z_0 = [0.0000, 0.0000, 1.0000]T

J_v2 = z_1 × (o_4 - o_1) = [-0.4899, -0.2828, 0.4243]T
J_w2 = z_1 = [0.5000, -0.8660, 0.0000]T

J_v3 = z_2 = [0.5000, -0.8660, 0.0000]T
J_w3 = [0.0000, 0.0000, 0.0000]T

J_v4 = z_3 × (o_4 - o_3) = [-0.1561, 0.0254, -0.1225]T
J_w4 = z_3 = [-0.6124, -0.3536, 0.7071]T

J =
[-0.3456 -0.4899   0.5000 -0.1561]
[0.7522 -0.2828 -0.8660   0.0254]
[0.0000   0.4243   0.0000 -0.1225]
[0.0000   0.5000   0.0000 -0.6124]
[0.0000 -0.8660   0.0000 -0.3536]
[1.0000   0.0000   0.0000   0.7071]

J matches the code output.

Part 3: Group Specific Task

Output:-

```
Converged in 10 iterations!
Final Joint Angles: q1 = -135.00°, q2 = 126.38° q3 = -152.76° q4 = 179.99° q5 = 153.62° q6 = 134.99°
```

Out of working range:

```
Did not converge in 209 iterations.
Final Joint Angles: q1 = -99.50°, q2 = 75.87°
```

Initial value made the number of iterations that were needed to converge. Singularities were not observed.

Video attached for demonstration.